

transition from a given asymptotic cycle to only a few other asymptotic cycles when one node is flipped.

Kauffman argues that in real cells, similar global behavior is observed (see Figure 2.15.) For a given organism, there are only a limited number of cell types, generally fewer than one hundred. Also, cell cycles are short, of the order of several hours. This corresponds roughly to the cycle lengths found by Kauffman, if one assumes that the time between iterates is the time it takes to turn a gene on or off—roughly one minute. Further, in many organisms, cells have only a limited ability to differentiate. They can only differentiate into a limited number of cell types, and these transitions will occur only if the cell is in the “right” place at the “right” time.

The “short” cycles observed by Kauffman correspond to a huge amount of dynamical order being generated by the random Boolean network. For Kauffman’s networks with $K = 2$, the typical cycle has a length of roughly six time steps when $N = 200$. On the other hand, it has been found that for random Boolean networks where $K = N$, the cycle times are huge, roughly $2^{\frac{N}{2}}$. For $N = 200$ this is ten million times the estimated age of the universe (assuming 10^{-6} sec per transition). Thus, it seems that the specific N -input Boolean functions that arise from networks with $K = 2$ inputs produce much shorter cycles than randomly selected Boolean functions.

2.5 CELLULAR AUTOMATA

Gene networks consist of many different genes, each of which may have its own distinct regulatory mechanism. In contrast, some systems consist of many copies of the same thing, each regulated by exactly the same mechanism. For example, this is the case of the cells in living organisms, each of which contains the same set of genetic instructions. It might be thought that in a network of similar elements, all elements will behave similarly. This is often far from the case, however.

Motivated by the examples of living organisms, scientists have formulated the concept of a **cellular automaton**. In a cellular automaton, there are identical elements, usually located in a regular array that can be one-dimensional, two-dimensional, or even higher-dimensional. The update rule for each “cell” depends on that cell and some of its nearest neighbors as shown in Figures 2.16 and 2.17.

For the geometries drawn in these figures, the nodes on the edge of the automaton do not have as many neighbors as the nodes in the center. In order to make all nodes the same, it is conventional to connect the nodes on the edge to one another, so that one-dimensional cellular automata have the topology of a ring, and two-dimensional cellular automata have the topology of a torus.

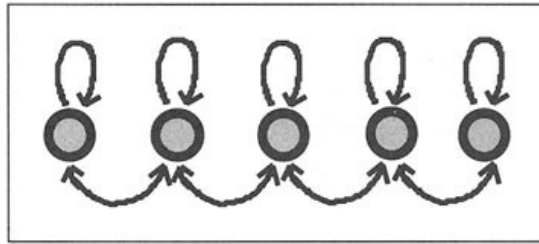


Figure 2.16 A geometry of a one-dimensional cellular automaton in which each cell has two neighbors. In some cases, a connection is made between the two cells on the ends in order to form a ring.

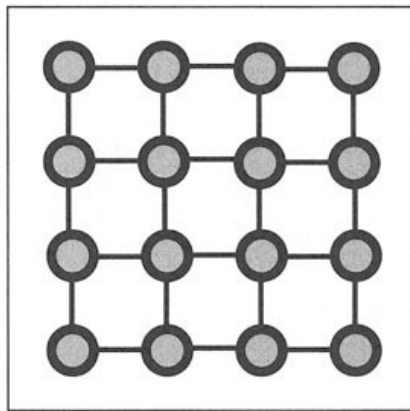


Figure 2.17

A geometry of a two-dimensional cellular automaton in which each cells has four neighbors. The cells on the edges can be connected to each other, forming a torus.

BOOLEAN CELLULAR AUTOMATA

In **Boolean cellular automata**, the state of each of the nodes is 0 or 1 and the nodes have a Boolean rule. One type of geometry for a one-dimensional cellular automaton has each node take input from itself and from its nearest neighbors on either side. Since there are three inputs, there are $2^3 = 256$ possible Boolean rules. These cellular automata can produce varied types of behavior, depending on the rule and the initial condition. For example, the rule $(111) \rightarrow 0$, $(110) \rightarrow 0$, $(101) \rightarrow 0$, $(100) \rightarrow 0$, $(011) \rightarrow 1$, $(010) \rightarrow 0$, $(001) \rightarrow 1$, $(000) \rightarrow 0$ (which is conventionally named rule 10 because the binary digit 00001010 equals ten) produces left drifting stripes. In a “movie” of the automaton’s activity, which Figure 2.18 depicts, these lines wrap around from one side to the other, as a result of the ring geometry. (The stripes can be either one or two nodes wide. Why?)

Not all rules produce such simple behavior. For example, rule 45 $((111) \rightarrow 0, (110) \rightarrow 0, (101) \rightarrow 1, (100) \rightarrow 0, (011) \rightarrow 1, (010) \rightarrow 1, (001) \rightarrow 0, (000) \rightarrow 1)$ produces a much more complicated pattern, as Figure 2.19 displays.

Another one-dimensional cellular automaton that produces interesting patterns is rule 90: $((111) \rightarrow 0, (110) \rightarrow 1, (101) \rightarrow 0, (100) \rightarrow 1, (011) \rightarrow 1, (010) \rightarrow 0, (001) \rightarrow 1, (000) \rightarrow 0)$. Figure 2.20 shows the pattern gener-

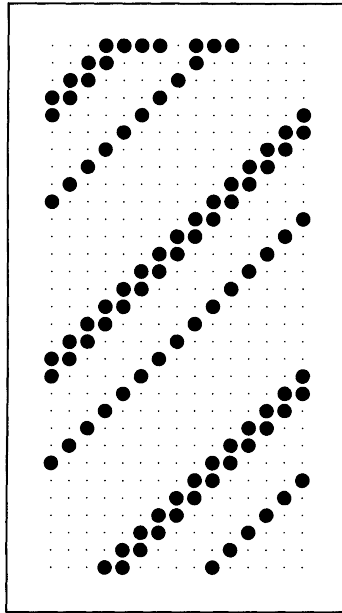


Figure 2.18

Activity of rule 10 in a cellular automaton with fifteen cells in a ring topology.

ated using rule 90 from a randomly selected initial condition. Such patterns can also be found in nature: A photo of a *conus* sea shell shows a strikingly similar pigmentation (see Figure 2.21).

Although one should not presume that this superficial resemblance implies that anything controlling pattern formation results in seashells, it is remarkable that such subtle patterns can be generated using simple algorithms. We will return to this subject further in Chapter 3.

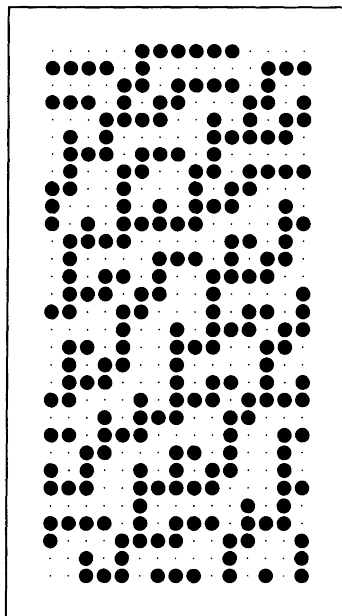


Figure 2.19

Activity of rule 45 for fifteen cells in a ring topology.

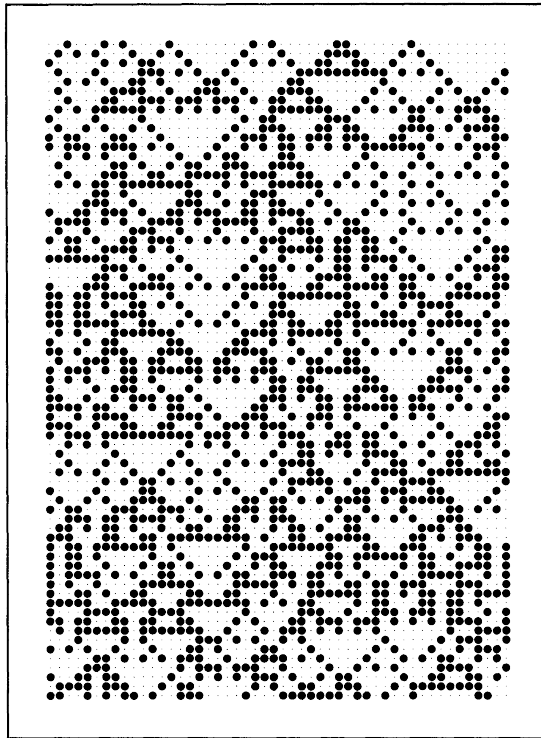


Figure 2.20
Activity of rule 90.

It is possible, of course, to have Boolean cellular automata with more than three inputs to each node. An early cellular automaton was the **game of “Life”** invented by the British mathematician John Conway in the 1970s. Conway imagined a two-dimensional geometry where each node is connected to itself and to its eight nearest neighbors. There are $2^{2^9} = 1.34 \times 10^{154}$ possible Boolean cellular automata with nine inputs, but Conway chose a particular rule inspired by interactions of living organisms with one another. Conway imagined that at any

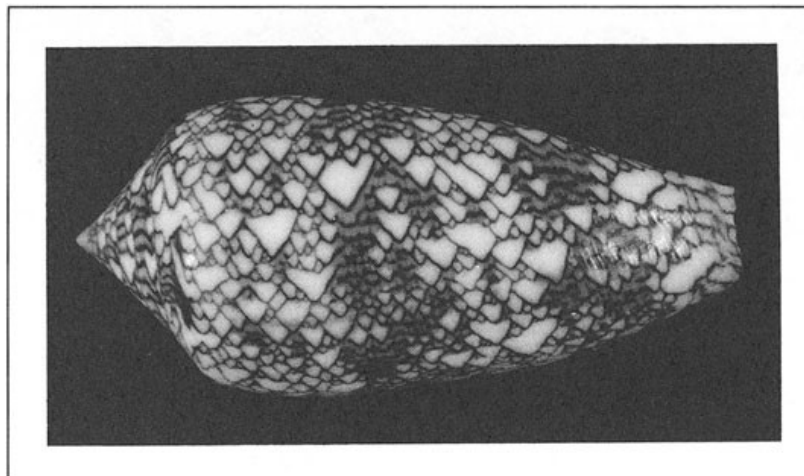


Figure 2.21 A conus sea shell.

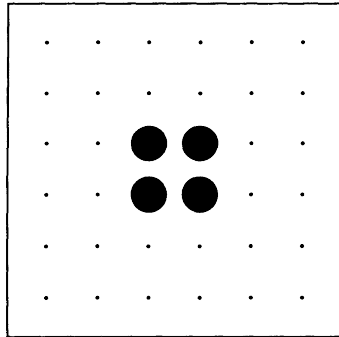


Figure 2.22
A fixed point in the “game of Life.”

time each node was either “alive” (1) or “dead” (0). In order for a living cell to stay alive at time $t + 1$, either two or three of its neighbors have to be alive at time t . The rationale for this is that if fewer than two neighbors are alive, then life cannot sustain itself. If more than three neighbors are alive, then there is overcrowding and once again the conditions for staying alive are not met. On the other hand, if a node is dead and four of its neighbors are alive, the node is “born” and becomes alive.

As whimsical as these rules may be, they do provide for a rich evolution starting from arbitrary initial conditions. Various forms of fixed points, cycles, and transients are possible. For example, an isolated square island of four living nodes is a fixed point (Figure 2.22), as is an isolated arrangement of eight living nodes surrounding a dead node. Periodic cycles are also possible: There is a cycle of period 2 involving a cluster of four and five cells (Figure 2.23).

NON-BOOLEAN CELLULAR AUTOMATA

Although we have just considered cellular automata in which the rules in each cell were Boolean functions, it is easy to imagine extending this concept so that variables may assume more than two values and other rules besides Boolean functions are allowed. For example, numerical schemes to solve partial differential

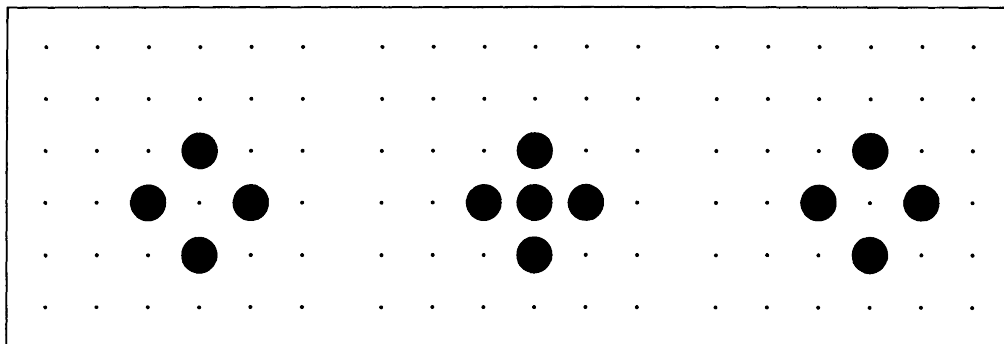


Figure 2.23 A cycle of period 2 in the “game of Life.”

equations that represent dynamics in space can be thought of in terms of cellular automata. Consider the problem of diffusion. If we allow the state of each element to be any real number, we can make a cellular automaton model of diffusion along a line. If $c_t[m]$ is the concentration of the substance at site m at time t , then we can write $c_{t+1}[m]$ as a function of $c_t[m - 1]$, $c_t[m]$, and $c_t[m + 1]$:

$$c_{t+1}[m] = D(c_t[m - 1] + c_t[m + 1]) + (1 - 2D)c_t[m].$$

D is the diffusion coefficient, which describes how fast the substance diffuses. The activity of this cellular automaton is shown in Figure 2.24, where the initial condition is that a single node has a nonzero concentration of the substance. As time progresses, the substance diffuses out to other nodes.

There are many other physical and biological phenomena that have been modeled with cellular automata, including the patterns of coloration in animals, the immune system, interaction of predators and prey, aggregation of particles, fluid flow, structure of the visual cortex, organization of ant trails, and growth of bacteria.

EXCITABLE MEDIA

One important application of cellular automata relates to **excitable media**. There are many systems in which a wave of some sort can pass through a medium, after which the medium cannot support another wave until a suitable length of time, called the **refractory time**, has passed. For example, if a wave of fire burns through a forest, there cannot be another forest fire until enough vegetation has grown to support a new fire. Media that display the joint properties of wave propagation and refractoriness are called **excitable media**. Other examples

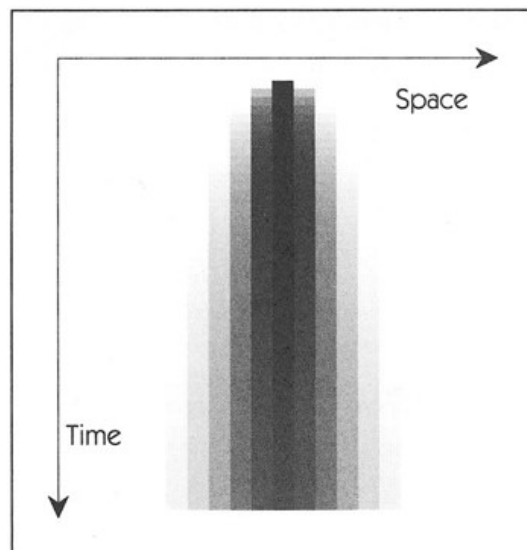


Figure 2.24
Diffusion in a cellular automaton.

include certain chemical media, such as the Belousov-Zhabotinsky reaction (see Figure 2.25) in which autocatalytic oxidation reactions take place, as well as neural and cardiac tissue.

If an excitable medium is two-dimensional (i.e., a sheet of medium, or a very thin layer), there are several fundamentally different geometries of waves that can be observed. In one, waves originate from a point source (e.g., the place where lightning started a forest fire) and spread in concentric circles. The effect is similar to ripples on a pond, except that two waves that meet will annihilate each other due to the refractory properties of the medium. Once the wave has moved off the edge of the medium, another wave will not occur unless it is started anew. An excitable medium—which is nonlinear—is fundamentally different than a linear medium. Waves in a linear medium, such as small waves on the surface of a pond, pass through one another, rather than annihilating each other.



Figure 2.25 Waves in the Belousov-Zhabotinsky reaction. Reprinted from Winfree (1975). Copyright 1975 by the AAAS.

For an appropriate geometry of an excitable medium, waves can also move in closed circuits. Imagine an excitable medium arranged in the form of a ring. If a wave that propagates in one direction is started, it will eventually come back to its starting place. If the circumference of the ring divided by the speed of the wave is longer than the refractory period of the medium, the wave will continue circulating around and around the ring. In a linear medium, waves can also travel along a ring. However, the amplitude of the wave will eventually diminish due to dissipation of energy, and the wave will die out. In an excitable medium, the wave may never die out.

Another geometry of waves involves propagation of spiral waves of activity (see Figure 2.12). Here the wave moves in a circuit even though there is no hole in the medium. Instead, the wave is formed into a spiral and the “hole” is a functional one created from moment to moment by areas of refractory medium. The spirals can be stable, or can meander in complex geometries, or might even break up into multiple spirals (see Figure 2.26). Such phenomena are believed to underlie pathological patterns of excitation in cardiac tissue, called **tachycardia** and **fibrillation**. In tachycardia, the heart beats much more rapidly than normal. In fibrillation, the heart does not contract in an organized way and is ineffective in pumping blood; this condition is the immediate cause of death in perhaps one in three people.

Cellular automata provide a mathematical model for excitable media. Let each node be in one of several states:

Q Quiescent but excitable. This is the state of those nodes that are not carrying a wave, but that could do so if stimulated.

E Excited. This is the state of those nodes the wave is passing through at the present instant.

R Refractory. This is the state of nodes that a wave has recently passed through, and which have not yet recovered enough to become excitable.

The rules of the automaton are simple:

- If a node is quiescent at time t , then it stays quiescent at time $t + 1$ unless one or more of its neighbors is excited at time t , in which case the node becomes excited at time $t + 1$.
- If a node is excited at time t , it becomes refractory at time $t + 1$.
- If a node is refractory at time t , it becomes quiescent at time $t + 1$. Actually, it is often desirable to have the refractory period last more than one time unit. If this is the case, then when the node first becomes refractory it is assigned a value $r = \mathcal{T}$. If the node has value r at time t , then its value becomes $r - 1$ at time $t + 1$. When $r = 0$, the cell is quiescent. $\mathcal{T}$ is the refractory period.

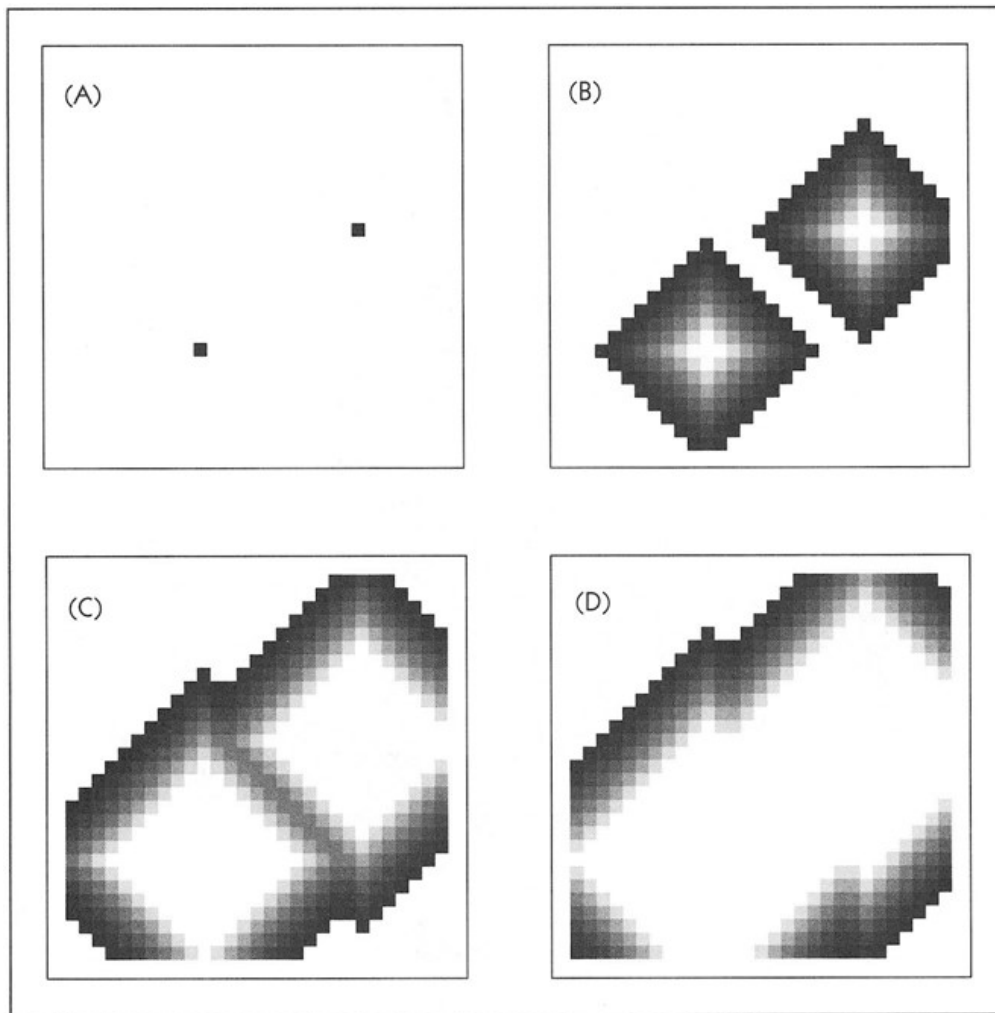


Figure 2.26 Waves in an excitable medium. The excited nodes are shown in black, the refractory nodes in shades of gray, and the quiescent nodes in white. Each node gets inputs from itself and from its four nearest neighbors. (A) Time $i = 0$. The medium is stimulated at two points to start waves. (B) Time $i = 8$. The waves spread out from each point. (C) Time $i = 14$. When the wavefronts touch, they annihilate each other. (D) Time $i = 17$. The waves will spread out until the whole medium becomes quiescent. No new wave will start until the medium is stimulated again.

These rules for cellular automata not only generate spreading waves, as seen in Figure 2.26, but also spiral waves as illustrated in Figure 2.27.

These rules can be modified to correspond to specific physical situations—for example, in heart tissue the refractory period can be made to depend on the time between excitations—but the essential rule for excitable media is simple: quiescent until excited, then refractory, then quiescent again.

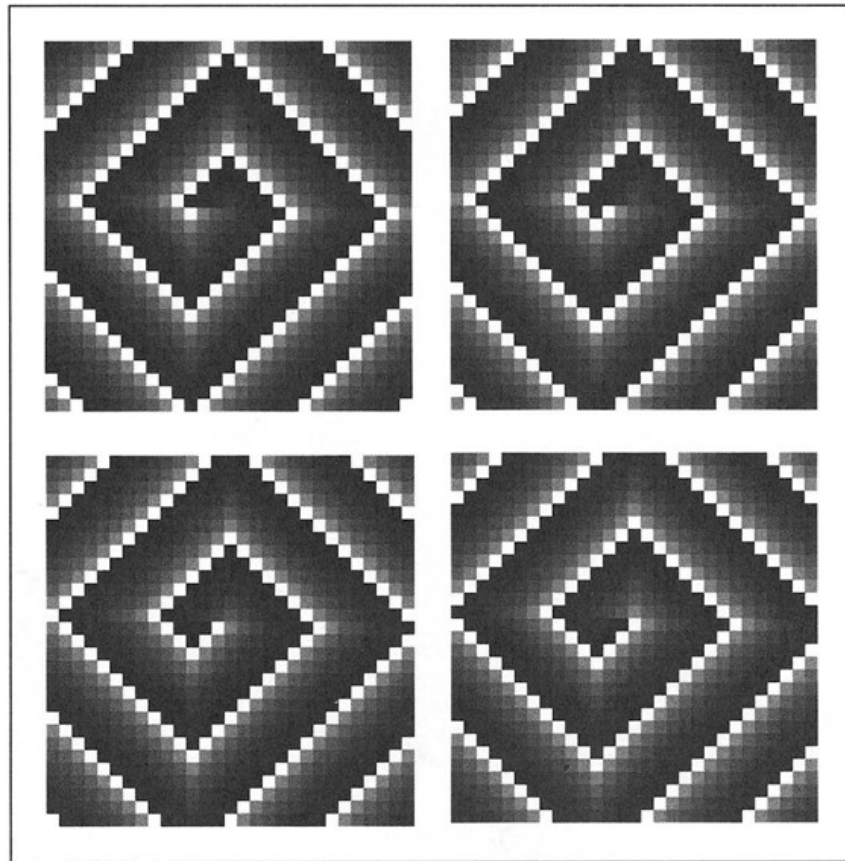


Figure 2.27 The state of the network in four successive times in a spiral wave in an excitable medium. The wave is rotating counterclockwise.

DYNAMICS IN ACTION

4 SPIRAL WAVES IN CHEMISTRY AND BIOLOGY

The study of excitable media has been one of the triumphs of applications of non-linear dynamics. For a detailed accounting, see Winfree (1987). We briefly mention two different aspects of excitable media: oscillations and waves in chemistry, and oscillations and waves in biological systems.

The earliest published report of an oscillating chemical reaction that we are aware of was published by W. C. Bray in 1921. This reaction uses very common ingredients that are easy to find. Bray's recipe is shown in the following table. Prepare a mixture at the concentrations shown. Heat to 60° C. If you follow this recipe you should be able to observe spatially homogeneous oscillations in which there is a slow buildup

of a brownish color (I_2) and a rapid disappearance. It takes a bit of a time for the oscillation to set in, so be patient. We can only speculate whether periodic acid (Cotton and Wilkinson, 1980) is an intermediate in this reaction.

H_2O_2	0.19 M
KIO_3	.094 M
H_2SO_4	.073–.0961 N

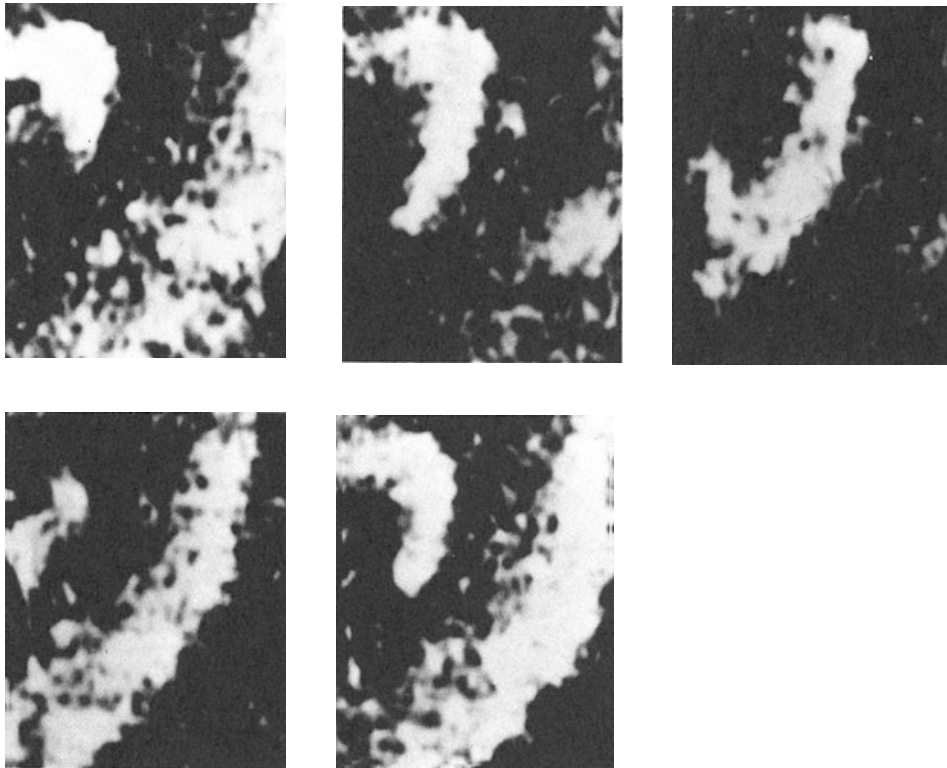
The interest in oscillating chemical reactions and excitable media was stimulated in the early 1970s as news of the Belousov-Zhabotinsky reaction spread. A recipe for this reaction was given by Winfree (1972, 1980). Here is the recipe, but consult the original sources for a couple of tricks.

To 67 ml of water, add 2 ml of concentrated sulfuric acid and 5 gm of sodium bromate (total 70 ml). To 6 ml of this in a glass vessel, add 1 ml of malonic acid solution (1 g per 10 ml). Add 0.5 ml of sodium bromide solution (1 g in 10 ml) and wait for the bromine color to vanish. Add 1 ml of 25 mM phenanthroline ferrous sulfate and a drop of Triton X-100 surfactant solution (1g in 1000 ml) to facilitate spreading. Mix well, pour into a covered 90 ml Petri dish illuminated from below.

Once the reaction gets started you should see blue rings propagating from localized regions on a red background. If you give the dish a gentle shake to break up the rings, you will see spectacular geometries similar to those shown in Figure 2.25.

The observation of spiral geometries in living systems has important implications for human health. In the normal heart, a specialized region—the sinoatrial node—sets the frequency. However, in some dangerous cardiac arrhythmias, the frequency of the oscillation is set by the time it takes a wave of cardiac excitation to travel in a circuitous pathway. This type of mechanism is believed to underlie potentially fatal cardiac arrhythmias such as ventricular fibrillation. In order to record the activation of cardiac tissue in space and time, a new methodology has been developed in the past decade that promises to revolutionize the study of waves in excitable media, by using fluorescent dyes that respond to local changes in membrane potential or chemical concentrations. A dramatic demonstration of reentrant waves in cardiac tissue was given by J. M. Davidenko and coworkers (1992), who studied this type of

wave in a piece of cardiac muscle in vitro (see figure on the next page). Spiral waves have been found in other biological systems, including aggregation of slime molds, where they are due to chemotactic mechanisms relying on pulsatile secretion of cyclic adenosine monophosphate (AMP). Spiral waves have also been associated with fluctuations of Ca^{2+} in amphibian oocytes.



Spiral waves in cardiac tissue. From Davidenko et al. (1992). Reprinted with permission from *Nature*. Copyright 1992 Macmillan Magazines Limited.

Thus, spiral waves of activity can be observed in a wide range of different systems. These waves can be a chemical curiosity, a life-threatening condition, a vital step in the life cycle of amoeba, or of still-unknown significance (the Ca^{2+} waves). In each of these, the underlying physical and chemical mechanisms are certainly different. Yet, spiral waves emerge from the basic mechanism of wave propagation in excitable media that we have discussed and modeled in a simple way in the cellular automata. Therefore, in understanding spiral waves, it is as important to understand the mathematics of excitable media as it is to understand the specific

mechanisms of halogen chemistry, the ionics of cardiac propagation, or cyclic AMP or Ca^{2+} metabolism.

2.6 ADVANCED TOPIC: EVOLUTION AND COMPUTATION

In the networks considered so far, the structure does not change in time. However, in many biological situations network structure and connectivities do change in time or **evolve**. Many unsolved problems in science involve understanding the mechanisms by which networks can change their structure.

In a biological organism, the inherited instructions are contained in the genes, and are called the **genotype**. The term **phenotype** refers to the visible properties of the organism as determined by the genotype and the environment.

Classical Darwinian evolution assumes that there is random phenotypic variation combined with selection of the fittest. Although Darwin did not know about genes, we now recognize that the phenotype is an expression of the genotype. Natural populations consist of organisms with different genotypes. The relative prevalence of a given genotype depends on the success of reproducing—the fitness—of the corresponding phenotype in preceding generations. There are two different types of mutations, point mutations and mutations in which segments of DNA recombine.

John Holland has used the term **genetic algorithm** to refer a vastly simplified computer simulation of biological evolution. Nevertheless, genetic algorithms implemented on computers preserve the following essential features of biological evolution: a population of genotypes, reproduction with mutation, and selection according to the phenotype. The presence of recombination distinguishes genetic algorithms from other types of search optimization algorithms.

In order to study the evolution of dynamical systems, investigators have studied cellular automata whose rules evolve in time. Since for a cellular automaton, the same Boolean rule is applied to every cell, we can consider the genome of a cellular automaton to be the sequence of ones and zeros in the output of the Boolean rule. A point mutation is caused when a 1 is randomly changed to a 0, and a recombination occurs when some of the ones and zeros from one rule are replaced with the ones and zeros from another. This provides all of the features for evolution, except for selection.

In nature, organisms are selected based on the ability of their phenotype to survive and reproduce in their environment. For cellular automata, we impose a selection criterion based on the performance of a computational task. One task that has been studied is the ability of a cellular automaton to perform the following computation—the determination of whether 50 percent or more of the cells in an initial condition are ON. If more than 50 percent of the cells are ON, the cellular automaton should approach a steady state in which all cells are ON. Otherwise it should approach a steady state in which all cells are OFF. If the cellular automaton has only a small local neighborhood, it is not known which cellular automaton can perform the task the best.

Norman Packard studied this problem for a cellular automaton in a class of 1-dimensional Boolean cellular automata in which the state of each element depends on itself and on its three nearest neighbors on each side—7 inputs altogether. Since there are $2^7 = 128$ different states of the inputs for each cell, there are $2^{2^7} = 2^{128}$ different cellular automata that can be constructed. This is an astronomically large number, and consequently it is impossible to study the dynamics of each member of this class. Since there is no theory that allows one to predict the dynamics of any given network without actually simulating it, it is necessary to search the space of different cellular automata.

Packard (1988) approached the problem using genetic algorithms. In the first generation a population of randomly generated cellular automata were iterated for many time steps for each of several different initial conditions. Each cellular automaton was rated based on its ability to carry out the assigned computational task. Then, in analogy to biological evolution, those cellular automata in the population that performed best were allowed to “reproduce” for the next generation. In addition, the cellular automata were allowed to interchange bits and pieces of their truth tables, and to mutate by changing entries in their truth tables. This yields a new population of cellular automata—generation 2. Once again each cellular automation in generation 2 was tested against a set of new initial conditions. As before, the cellular automata reproduced and mutated. The process was continued for many generations.

Packard’s work was preliminary and was subsequently redone and greatly extended by James Crutchfield, Raja Das, Peter Hraber, and Melanie Mitchell. Here, we present the most interesting results from the computer experiments performed by Das, Mitchell, and Crutchfield (1994). Figure 2.28 shows an illustration of the best cellular automation that was found in generation 12. About 46 percent of the cells are OFF in the initial state and after 148 iterations, all the cells are ON. Therefore, this cellular automaton has incorrectly classified the original distribution (though on many other initial conditions it comes up with the correct classification). This cellular automaton had a simple strategy for classifying figures. Unless there are large blocks that are all OFF cells, it will map to a state

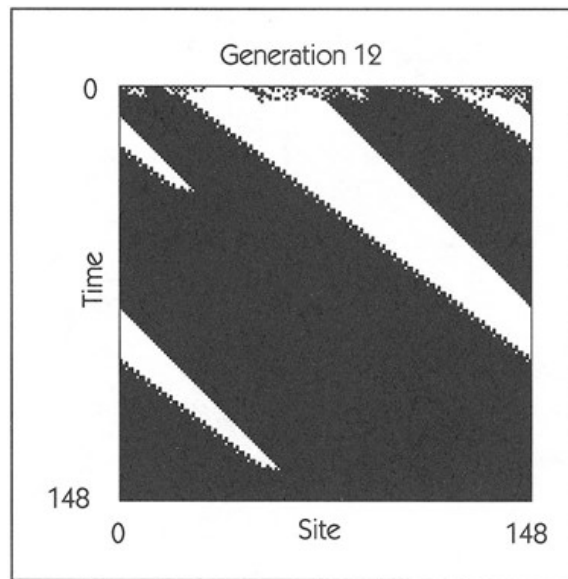


Figure 2.28

A cellular automaton that has evolved under selective pressure to perform the computational task of deciding if there are more ON or OFF cells at time zero. This one makes the wrong decision. Figure provided by James P. Crutchfield based on results in Crutchfield and Mitchell (1994) and Das, Mitchell, and Crutchfield (1994).

of all ON cells. Figure 2.29 shows the operation of the best cellular automaton in generation 18. Starting from the same initial condition, this cellular automaton correctly classifies the initial condition since all the cells converge to a OFF state. This system has developed a more sophisticated strategy for classifying the initial configuration based on transmitting information to make logical decisions about the initial condition. This is seen in the figure as the emergence over time of particles and particle interactions. The particles appear as boundaries between homogeneous regions. Their propagation and subsequent interaction implement

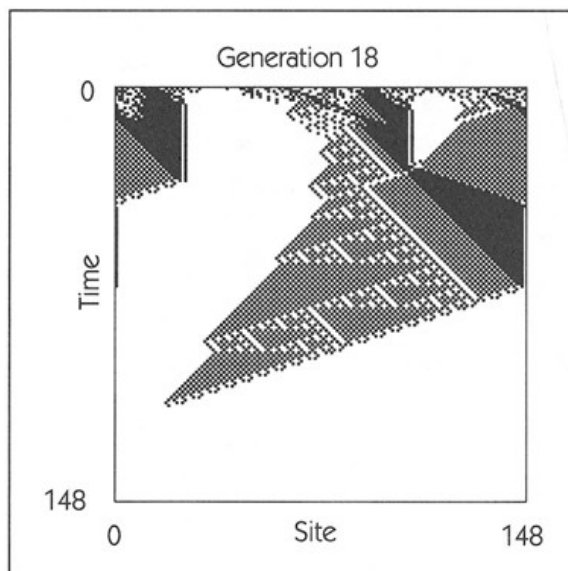


Figure 2.29

The best cellular automaton at generation 18 with the same initial condition as in Figure 2.28. This cellular automaton successfully classifies the initial condition and converges to a state in which all cells are OFF. Figure provided by James P. Crutchfield based on results in Crutchfield and Mitchell (1994) and Das, Mitchell, and Crutchfield (1994).

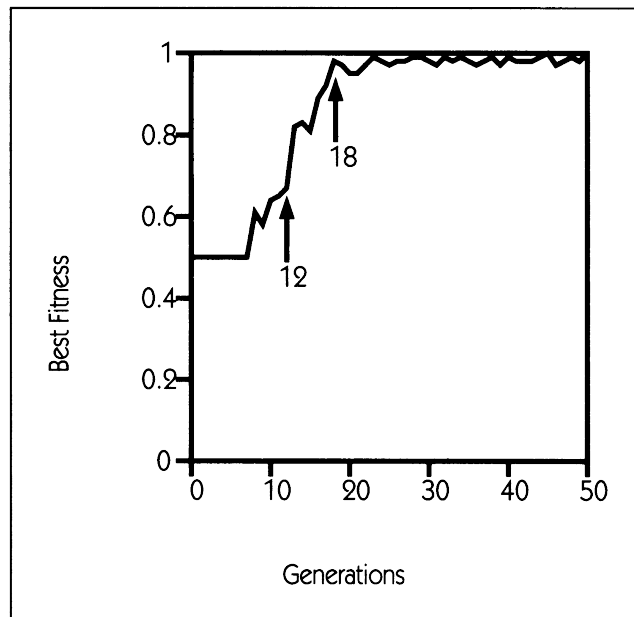


Figure 2.30 Best fitness in a population of cellular automata as a function of generation number. Generations 12 and 18 are marked with arrows. Figure provided by James P. Crutchfield based on results in Crutchfield and Mitchell (1994) and Das, Mitchell, and Crutchfield (1994).

the new computational strategy: Various logical decisions are made about large regions in which the initial condition's density was ambiguous.

What is interesting here is that over several generations, the genetic algorithm leads to an evolution in the truth tables so that the performance on this simple computational task improves over time. Figure 2.30 illustrates this evolution showing the increase of the best fitness over several generations. This is an important finding. It shows that one can simulate evolution in simple model settings. To figure out what is happening, we have to think about what strategy (computation) the cellular automata have been evolved to adopt in order to “survive”—almost as though they are alive. Thus, these sorts of simple models may provide the sorts of “data” that will help researchers formulate rules for evolution in complex systems.

SOURCES AND NOTES

To obtain an introduction to the mathematics of Boolean algebra see (Hohn, 1966).

Jacob and Monod (1961) and Monod *et al.* (1965) present early discussions of the control of cellular metabolism by gene networks. More recent texts provide introductions to genetic enzymatic regulatory mechanisms (Alberts, 1983;

Ptashne, 1986). Kauffman's (1969) original exposition of the properties of his model gene networks presents a provocative notion of how genes can be organized. The flowering of these ideas over the past quarter of a century is recounted in (Kauffman, 1993).

The analysis of excitable media is a very active area of current research with implications in physics, chemistry, and biology. Winfree (1980) gives some hints on how to get the Belousov-Zhabotinsky reaction to run (p. 301). Winfree (1987) presents a good introduction to the field with extensive references to problems in cardiac arrhythmias, aggregation of slime molds, and chemical oscillations. Gerhardt *et al.* (1990) demonstrated that many of the subtle properties of excitable media can be captured by cellular automata models and provide references to earlier studies of cellular automata models of excitable media. Experimental observation of spiral waves in cardiac tissue was carried out by Davidenko *et al.* (1992). Spiral waves of Ca^{2+} in oocytes were photographed by Lechleiter *et al.* (1991).

A discussion of logical models for lambda bacteriophage is given by Thomas, Gathoye, and Lambert (1976) and Thieffry and Thomas (1994). Thomas has been particularly interested in logical models for a variety of systems as reviewed in the monograph by Thomas and D'Ari (1990). A review of biological application of cellular automata models in diverse applications can be found in Ermentrout and Edelstein-Keshet (1993).

The study of computation and evolution has generated a significant interest among scientists and in the popular press. An important contribution was John Holland's (1975) formulation of the concept of genetic algorithms. Thinking about cellular automata from the perspective of mathematics and physics is ably reviewed by Stephen Wolfram (1983, 1984), who is better known as the guiding force behind the development of Mathematica®. More recently, Packard (1988), Kauffman (1993) and others at the Santa Fe Institute in New Mexico have been pursuing the hypothesis that a simple generalization will be found that will have predictive value in explaining evolution in diverse settings. The hope is that a "law" will be found for evolution in complex adaptive systems of similar power to the second law of thermodynamics in closed chemical systems. Packard interpreted his results as indicating that on the computational task described in the text, there was evolution to the "edge of chaos". In another context, Crutchfield and Young (1989) analyzed computation at the "onset of chaos". Several recent popularizations (Waldrop, 1992; Lewin, 1992) have focused on the theme of evolution in complex systems as a central theme.

In their careful studies, Mitchell, Hraber, and Crutchfield (1993) failed to reproduce the aspects of the original results of Packard that supported the argument that there was evolution to the "edge of chaos". They also gave theoretical arguments as to why Packard's results could not support the "edge of chaos" claim. Finally, they note that in finite state systems, there is no clear definition of "chaos" or the "edge of chaos".

Nevertheless, Mitchell et al. (1993) and Das et al. (1994) concluded that the conception of using genetic algorithms to evolve computation in cellular automata is an important idea, and they have made significant advances in understanding the evolution of computational strategies in cellular automata.

EXERCISES

 **2.1** Explain the English-language names given to the Boolean functions of two inputs in Table 2.1. (HINT: The X in XOR stands for “exclusive.” The N in NAND and NOR means “not.” XNOR might make more sense if it were written NXOR, but that would be too hard to pronounce.)

 **2.2** Show that the network in Figure 2.31 and the following truth table are equivalent:

$(xy)_t$	$(xy)_{t+1}$	$(xy)_t$	$(xy)_{t+1}$
(00)	(10)	(10)	(11)
(01)	(00)	(11)	(01)

If the initial state at $t = 0$ is (00), then the network cycles through the states $(00) \rightarrow (10) \rightarrow (11) \rightarrow (01) \rightarrow (00) \dots$. This network is frustrated, and the cycle of length 4 is the only possible activity for the network. Why?

Write down the truth table for the three-node inverting network shown in Figure 2.32, and determine all modes of behavior starting from any initial state. Is this network frustrated? What is the longest possible cycle in this network? How many possible states of the network are there? Show that there must be more than one cycle or fixed point.

 **2.3** Consider a single-input Boolean network with n components arranged in a loop. The nodes will be numbered $1, 2, \dots, n$. We will write the state of

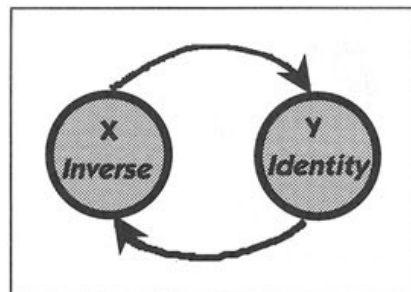


Figure 2.31

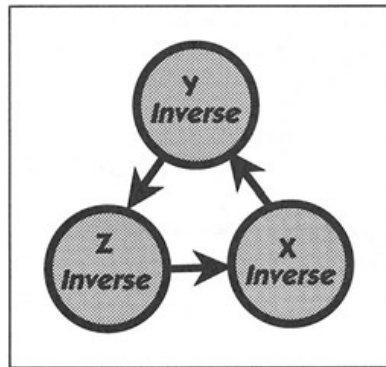


Figure 2.32

node k at time t as $x_t(k)$. The input to node 1 is node n . For all other nodes $k = 2, 3, \dots, n$ the input comes from node $(k - 1)$. All the nodes follow the IDENTITY rule, except for node n , which follows the INVERSE rule. This logical organization is a model for feedback inhibition, in which an end product can act to inhibit the production of an initial product. Start from an initial condition with $x_0(1)=1$ and $x_0(k) = 0$ for $k = 2, 3, \dots, n$.

- What are the dynamics when $n = 2$?
- What are the dynamics when $n = 3$?
- Generalize this to all values of n .

 **2.4** The neural network shown in Figure 2.33 has been proposed as a model of a biological oscillator. The symbol $\neg$ represents an inhibitory synapse and the symbol $\rightarrow$ represents an excitatory synapse.

Note that neuron 1 receives inhibitory inputs from neurons 3 and 4; neuron 2 receives excitatory inputs from neurons 1 and 3 and inhibitory input from

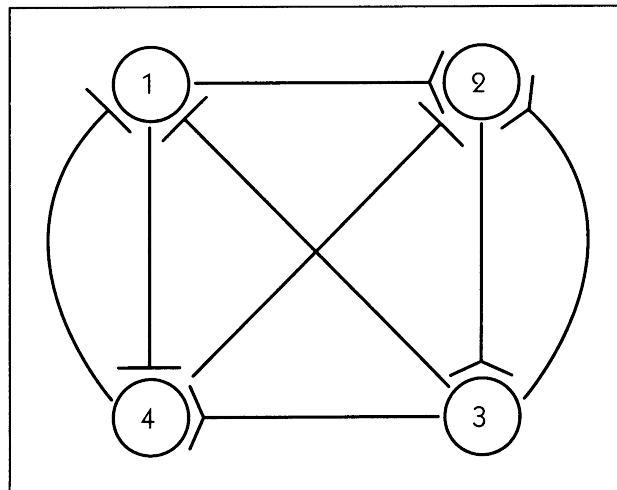


Figure 2.33

neuron 4, and so on. The truth table that relates the activity in this network at time t to the activity at time $t + 1$ is shown below.

State at time t	State at time $t + 1$	State at time t	State at time $t + 1$
(0000)	(1000)	(1000)	(1100)
(0001)	(0000)	(1001)	(0000)
(0010)	(0101)	(1010)	(0100)
(0011)	(0001)	(1011)	(0000)
(0100)	(1010)	(1100)	(1110)
(0101)	(0010)	(1101)	(0010)
(0110)	(0111)	(1110)	(0110)
(0111)	(0011)	(1111)	(0010)

- For each neuron, write the truth table for that neuron as a function of its inputs only.
- Assume that at $t = 0$ the state is (1000). Determine all future states until a steady state or cycle is reached.

 **2.5** Consider a one-dimensional Boolean cellular automaton, where each cell takes input from its two neighboring cells but not from itself. The Boolean rule is

Input at time t	Output at time $t + 1$
(00)	0
(01)	1
(10)	1
(11)	0

Let there be ten cells in the network.

- Assume that at time $t = 0$, the state in three cells (of your choosing) is equal to 1. The rest are 0. In the chart below, follow the evolution as time proceeds for five iterations using the scheme shown below. (Remember that cell 1 is connected to cell 10, and vice versa.)

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
$t = 0$										
$t = 1$										
$t = 2$										
$t = 3$										
$t = 4$										
$t = 5$										

Each square in your 10×6 array will have a 1 or a 0.

- Consider the evolution of the system described above in the limit $t \rightarrow \infty$, but starting from any arbitrary initial condition at $t = 0$. Does an initial condition exist which will give rise to (i) a steady state, (ii) a cycle, (iii) or chaos (i.e., aperiodic dynamics)?
- Give an upper limit (if one exists) for the length of a cycle in this system.

 **2.6** The following is a random Boolean switching network with three variables x , y , and z :

$(yx)_t$	z_{t+1}	$(yz)_t$	x_{t+1}	$(zx)_t$	y_{t+1}
(1 1)	0	(1 1)	0	(1 1)	1
(1 0)	0	(1 0)	0	(1 0)	0
(0 1)	0	(0 1)	0	(0 1)	1
(0 0)	1	(0 0)	1	(0 0)	0

- Fill in the following table:

$(xyz)_t$	$(xyz)_{t+1}$	$(xyz)_t$	$(xyz)_{t+1}$
(111)		(011)	
(110)		(010)	
(101)		(001)	
(100)		(000)	

- Determine the dynamics starting from all initial conditions.
- How many steady states and cycles are there in this network?

 **2.7** Consider a two-dimensional 5×5 Boolean cellular automaton. The neighbors of any given cell are the four cells that share a common edge with that cell. The rule for evolution is as follows: The state of a given cell will be 1 at time $t + 1$ if the state of that cell or the state of any of its neighbors was 1 at time t . Otherwise the state of the cell will be 0 at time $t + 1$.

- Choose three of the cells randomly and assume that the state of those cells is 1 at $t = 0$. Assume that the states of all the other cells in the network is 0 at $t = 0$. Draw a 5×5 grid and show the states of each of the twenty-five cells at $t = 0$.
- Show the subsequent evolution of this system. At each time step, show the state of each cell (either 1 or 0) until either a steady state or a cycle is reached.
- How many different ways could you have chosen the initial configuration in part a?
- Find an initial condition with three cells in state 1 that gives the shortest transient until a steady state or a cycle is reached. Show this configuration, and give state the length of the transient.

 **2.8** A Kauffman network contains four elements w, x, y , and z where the activity of each element can be represented by a Boolean switching network.

$(yx)_t$	w_{t+1}	$(wz)_t$	x_{t+1}	$(xw)_t$	y_{t+1}	$(wy)_t$	z_{t+1}
(00)	0	(00)	1	(00)	1	(00)	1
(01)	1	(01)	0	(01)	0	(01)	1
(10)	1	(10)	0	(10)	0	(10)	0
(11)	0	(11)	1	(11)	0	(11)	1

- Write a table that shows the state at time $t + 1$ given the state at time t for each of the sixteen possible states of the network.
- How many fixed points and cycles are there in this network? What are they?

 **2.9** A Boolean switching network has three elements. The network displays the following transitions:

$$001 \rightarrow 011 \rightarrow 111 \rightarrow 110 \rightarrow 100 \rightarrow 000 \rightarrow 001 \rightarrow \dots$$

$$010 \rightarrow 101 \rightarrow 010 \rightarrow \dots$$

- a. Write a truth table that gives the state of the network at time $t + 1$ for each of the possible states at time t .
- b. For each element of the network, indicate the inputs and the truth table that gives the output of that element as a function of the inputs.
- c. Out of the 4096 different “Kauffman-type networks” with three elements and no self-input, how many will have the same dynamical behavior? Justify your answer. By “same dynamical behavior” we mean networks with two cycles, one with six states and the other with two states. In the cycle with six states, each state is different from the preceding state in one node only. The cycle of two states flips between two states that differ in all three nodes. HINT: This requires some thought. There are a couple of ways to get the answer. You might wish to think about different ways of generating frustrated networks with three elements.

 **2.10** What does it mean to pick a “random” Boolean function? A Boolean function of k inputs gives the output for any given combination of ones and zeros. For example, one of the Boolean functions of two inputs, AND, is $(00) \rightarrow 0$, $(01) \rightarrow 1$, $(10) \rightarrow 0$, and $(11) \rightarrow 1$. We know that for two inputs there are four possible input states (00) , (01) , (10) , (11) . If we adopt the convention that the input states will always be written in increasing order (in base 2), we can make a shorthand notation for this Boolean function of (0001) , which reflects the output of the function for each of the four possible input states. Thus, the sixteen possible Boolean functions of two inputs can be written as

0000, 0001, 0010, 0011, $\dots$, 1110, 1111.

If we read this in base 2, then these sixteen Boolean functions are just 0, 1, 2, 3, $\dots$, 14, 15. So, each possible Boolean function of two inputs has a unique identifying number.

Consider Boolean functions of k inputs, and describe how to generate the unique identifying number for them.

In studying random Boolean networks, we may not want to use the Boolean functions CONTRADICTION and TAUTOLOGY. Show that for Boolean functions of k inputs, these two functions have the identifying numbers 0 and $2^{2^k} - 1$, respectively.

COMPUTER PROJECTS

Project 1 Write a program to generate random Boolean switching networks where each element receives 2 inputs. Restrict yourself to networks in which there is no self input. Starting from an initial condition iterate the network until a steady

state or cycle is reached. Your program should allow you to change the number of elements in the network.

Project 2 Generate 100 different random networks with 10 elements. For each network count the number of different cycles that you find starting from different initial conditions. It is necessary to test a large number of different initial conditions to guarantee that you have found all possible attractors. Find the mean number and standard deviation of the number of attractors per network.

Project 3 Now consider the influence of changing the number of elements of the network. As the number of elements increases, it is possible that the cycle can get astronomically long (but this does not happen too often). You will want to increase the number of elements in your network in a rather gradual manner, and figure out what to do if you never find a cycle. You might wish to consider the possibility of looking for cycles in a clever way. For example, suppose you just keep track of which elements change their states at each time step. What would happen if there is a periodic cycle? When Kauffman (1969) carried out this computation computers were rather primitive. You might wish to test your computing skills and your computers abilities to see how far up you can push the number of elements. Test to see if the cycle length and the number of cycles per network increase as the square root of the number of elements as was found by Kauffman.

Project 4 Generate random Boolean networks in which the number of inputs per element is different from 2. What happens to the cycle length and the number of cycles per network as a function of the size of the network?

Project 5 The game of “Life” and extensions provide students and recreational mathematicians with an interesting exercise: write a program to simulate the game of life. In doing this you should assume a geometry with cyclic boundary conditions. Generate random initial conditions and see what happens under subsequent iterations. How many different patterns can you find that represent repeating cycles?

Project 6 The following project is a difficult one, but would be a challenge to someone with interests in finite mathematics. Consider cellular automata that follow rule 90 with cyclic boundary conditions. Write a program to determine the number of different steady states and periodic orbits that are found for networks of different size as $t \rightarrow \infty$. In doing this you should try to take account of the cyclic symmetry so that two periodic cycles that only differ by their location on the ring are counted the same. Explore the way the number of attractors changes as the number of cells changes.

Project 7 Write a computer program to simulate wave propagation in excitable media. Vary the size of the neighborhood that can excite a given cell, the

number of inputs that have to be active in the neighborhood of a cell to excite the cell, and the duration of time each element is excited or refractory. Assign different colors to excited states, excitable states and refractory states. Try to find an initial condition that will evolve to spiral waves. One way to do this is to start from an initial condition in which a wave of excitation covers a some of the space, but a free end is left dangling into the rest of the media. How close can you come to simulating spiral waves such as are observed in the Belousov-Zhabotinsky reaction?